www.csvtuonline.com

Printed Pages : 3

Roll No. 3112413028

324731(24)

BE (7th Semester)
Examination, Nov.-Dec., 2016

(New Scheme)

Switchgear and Protection

Time Allowed : 3 hours

Maximum Marks : 80
Minimum Pass Marks : 28

Note : (*i*) Attempt **all** questions. Part (*a*) of each question is compulsory. Attempt any **two** parts from (*b*), (*c*) and (*d*), of each question.

(*ii*) The figures in the right-hand margin indicate marks.

1. (*a*) Why impedance relay is a voltage restrained over current relay? [2]

(*b*) Show the location of auxiliary contacts of C.B. in the trip circuit. [7]

(*c*) Explain the principle of instantaneous over current relay. How its characteristics is differed from I.D.M.T relay? [7]

(*d*) What is an impedance relay? Explain its principle of operation. Show its characteristics on R-X diagram. [7]

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

2. (*a*) What is magnetising inrush current? [2]

(*b*) Drawing neat circuit diagram, explain harmonic differential protection of transformer. [7]

(*c*) Using symmetrical component, determine currents in line of Y-Y earth transformer to L-G fault on a line. [7]

(*d*) A 6.6 kV, SMVA star-connected alternator has a reactance of 1.5 Ω/phase and negligible resistance. Merz-price protection scheme is used which operates when the out of balance current exceeds 25% of the full load corrents. The neutral of the generator is grounded through a resistance of 8 Ω. Determine the proportion of the winding which remains unprotected against earth fault. Show that the effects of the alternator reactance can be ignored. [7]

3. (*a*) What is meant by time graded protection? [2]

(*b*) What do you mean by power swing? Discuss the performance of MHO relay during subject→switchgear and protection power swing. [7]

(*c*) What is drawback of three-step distance protection? Explain how carrier-aided distance protection scheme gets rid of this drawback. [7]

(*d*) Draw phase comparison carrier current protection scheme in block form and explain the function of each block. [7]

www.csvtuonline.com

www.csvtuonline.com

4. (*a*) What is meant by duality related to comparators? [2]

(*b*) State the merits and demerits of static relay over electromagnetic relay. [7]

(*c*) What is phase comparator? Describe coincidence type phase comparator. [7]

(*d*) Draw the operating characteristic of distance relay used for the protection of medium length line. Design the relay using cosine phase comparator. [7]

5. (*a*) How does a circuit breaker differ from a switch? [2]

(*b*) Explain the principle of high resistance arc interruption in D.C. circuit. [7]

(*c*) Derive the expression for re-striking voltage and RRRV in terms of system voltage inductance and capacitance. [7]

(*d*) What is the resistance switching? Derive the expression for critical resistance in terms of system inductance and capacitance which give no transient oscillation. [7]

www.csvtuonline.com

www.csvtuonline.com